

DESIGN CRITERIA

	Heating	Cooling
Room set point	60.8 °F	78.8 °F
Outdoor dry bulb	23.2 °F	83.1 °F
Outdoor RH (coincident)	90 %	40 %
ROOM HUMIDITY		
Humidification set point	—	
lower RH limit		
Dehumidification set point	—	
upper RH limit		

SIZING (TAS DESIGN LOADS)

	Heating	Cooling
Design load	4,668 Btu/h	0 Btu/h
Design load per area	83.39 Btu/h·ft ²	0.00 Btu/h·ft ²
Sizing multiplier	not set	not set
Design loads: from Tas sizing — date/currency not recorded		

HEATING

Design day	London_Heathrow_Illustrative_Heating_Design_Day_99.6%_Coincident_Wet_Bulb_Extreme_Winter_Condition	
	Design day	Full year
Peak load	80,038 Btu/h	52,657 Btu/h
Peak hour	06:00–07:00	31 Dec 23:00–24:00
Room dry bulb	64.8 °F	64.4 °F
Room resultant	62.4 °F	62.8 °F
Room RH	23 %	45 %
Room humidity ratio	2.5 g/kg	5.1 g/kg
Outdoor dry bulb	—	19.8 °F
Outdoor RH	—	95 %

SENSIBLE COMPONENTS AT PEAK

Component	Design day (Btu/h)	Full year (Btu/h)
Lighting	0	1,194
Occupancy (sensible)	0	1,433
Equipment (sensible)	0	4,265
Infiltration / ventilation	-27,718	-20,515
Air movement (inter-zone)	-4,212	-3,370
Building heat transfer	-11,795	-8,004
External conduction — opaque	-23,165	-17,482
External conduction — glazing	-13,147	-10,179

LATENT COMPONENTS AT PEAK

Component	Design day (Btu/h)	Full year (Btu/h)
Occupancy (latent)	0	1,058
Equipment (latent)	0	154

Zero at both peaks (not listed): Solar, Air handling unit
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

COOLING

Design day

London_Heathrow_Illustrative_Cooling_Design_Day_0.4%_Dry_Bulb=>Mean_Coincident_Wet_Bulb_With_Global_Radiation

	Design day	Full year
Peak load	106.6 kBtu/h	98.2 kBtu/h
Peak hour	15:00–16:00	27 Jul 15:00–16:00
Room dry bulb	75.2 °F	75.2 °F
Room resultant	78.4 °F	77.4 °F
Room RH	55 %	52 %
Room humidity ratio	10.2 g/kg	9.8 g/kg
Outdoor dry bulb	—	86.2 °F
Outdoor RH	—	38 %

SENSIBLE COMPONENTS AT PEAK

Component	Design day (kBtu/h)	Full year (kBtu/h)
Solar	42.1	38.3
Lighting	7.2	7.2
Occupancy (sensible)	10.7	10.7
Equipment (sensible)	25.6	25.6
Infiltration / ventilation	4.2	3.4
Air movement (inter-zone)	1.6	1.2
Building heat transfer	6.8	5.0
External conduction — opaque	8.0	6.4
External conduction — glazing	4.2	3.8
Air handling unit	-3.8	-3.4

LATENT COMPONENTS AT PEAK

Component	Design day (kBtu/h)	Full year (kBtu/h)
Occupancy (latent)	7.8	7.8
Equipment (latent)	0.5	0.5

Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

RESULTS

	Heating	Cooling
Status	Available	Available
Result source	Tas	Tas
Read into model	27 Aug 2026 18:38	27 Aug 2026 18:38
Matches current model	not recorded	not recorded

Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year.
Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied.